

Lower lifetime dietary fiber intake is associated with carotid artery stiffness: the Amsterdam Growth and Health Longitudinal Study.

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BACKGROUND: Fiber intake is associated with lower cardiovascular disease risk. Whether arterial stiffness is influenced by lifetime fiber intake is not known. Any such association could explain, at least in part, the cardioprotective effects attributed to fiber intake. **OBJECTIVE:** The objective was to investigate whether a lower intake of fiber (and fiber-rich foods) throughout the course of young life (ie, from adolescence to adulthood) is associated with arterial stiffness in adulthood. **DESIGN:** This was a longitudinal cohort study among 373 participants in whom dietary intake was assessed between the ages of 13 to 36 y (2-8 repeated measures, median of 5), and arterial stiffness estimates of 3 large arteries (ultrasonography) were ascertained at age 36 y. **RESULTS:** After adjustment for sex, height, total energy intake, and other lifestyle variables, subjects with stiffer carotid arteries consumed less fiber (in g/d) during the 24-y study than did those with less stiff carotid arteries, as defined on the basis of the highest compared with the lowest sex-specific tertiles of the distensibility and compliance coefficients (reversed) and Young's elastic modulus: -1.9 (95% CI: -3.1, -0.7), -2.3 (-3.5, -1.1), and -1.3 (-2.5, -0.0), respectively. Furthermore, subjects with stiffer carotid arteries were characterized by a lower lifetime consumption of fruit, vegetables, and whole grains-deleterious associations that could be explained, to a great extent, by related low fiber intake. **CONCLUSIONS:** Lower lifetime intake of fiber during the course of young age is associated with carotid artery stiffness in adulthood. Promoting consumption of fiber-rich foods among the young may offer a means to prevent accelerated arterial stiffening in adulthood and related cardiovascular sequelae.